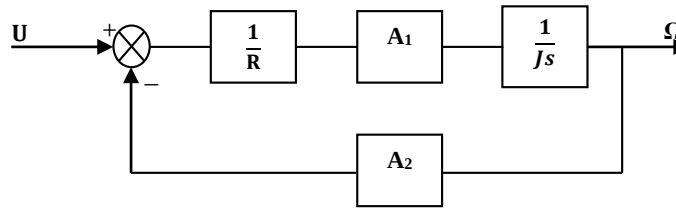


TD N°1

Exercice 1 :

The speed control of a DC motor is modeled by the following block diagram:



With: U armature voltage, Ω angular speed of the motor, A_1 : torque constant, A_2 : motor torque constant, J : total moment of inertia (mechanical load)

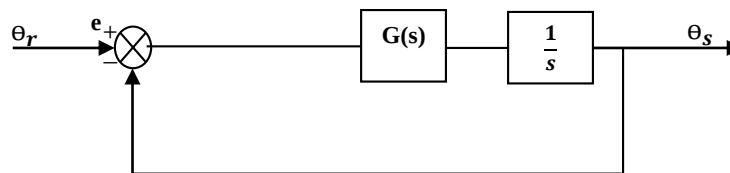
- 1) Derive the transfer function of this system $G(s) = \frac{\Omega(s)}{U(s)}$ and express it in the form of a first-order system:

$$\frac{K}{1+\zeta s}$$

Numerical application : $A_1 = 0.1 \text{ N.m/A}$, $A_2 = 0.1 \text{ V.s/rd}$, $R = 1 \text{ ohm}$ et $J = 5 \cdot 10^{-4}$.

- 2) A digital control loop is to be implemented, compute the sampling frequency required for this system.

Now, the same motor is controlled in position, as shown in the following block diagram:



With : θ_r angular position (reference), θ_s actual angular position (real output).

- 3) Derive the closed-loop transfer function: $F(s) = \frac{\theta_s(s)}{\theta_r(s)}$, express it in the canonical form of a second-order system.
- 4) Determine the **sampling frequency** required if the system is to be digitally controlled

Exercise 2 : Consider the system modeled by the following transfer function:

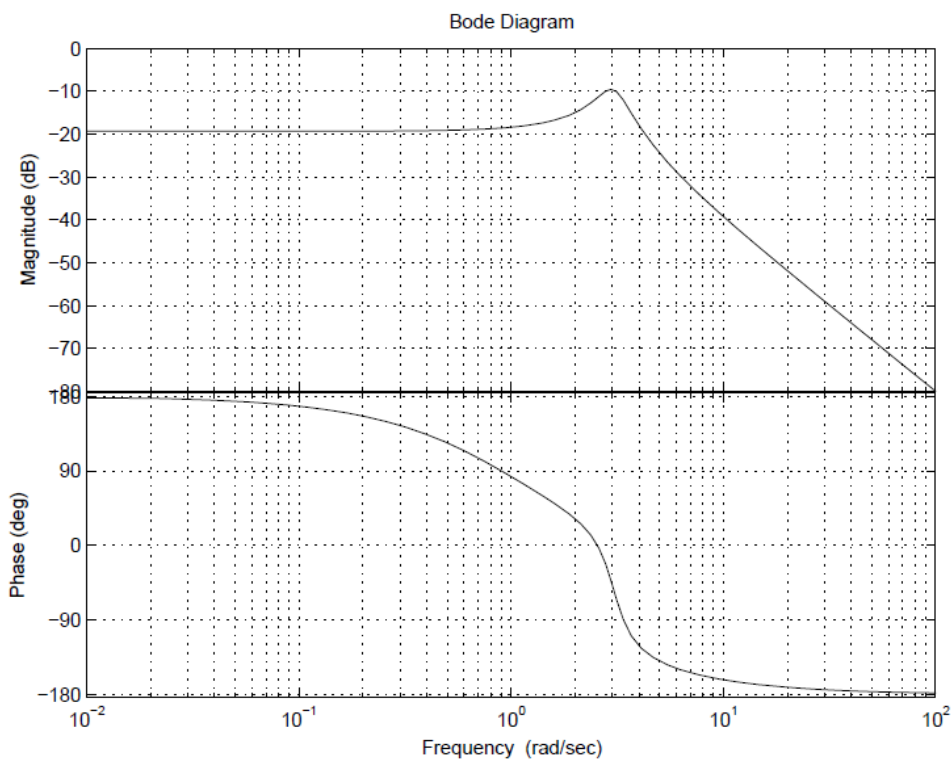
$$F(s) = \frac{10}{(s + a)(s + b)(s + c)}$$

With $a = 0.1$, $b = 0.5$, $c = 1$

- 1) What sampling period should be chosen if the system is to be processed digitally?
- 2) Same question for the following system: $G(s) = \frac{1}{0.2s^2 + 1}$

Exercise 3 :

Given the Bode plot of the system modeled by : $G(s) = \frac{s-1}{(s^3+2s^2+10.25s+9.25)}$



- Determine the suitable sampling frequency for digital processing.

Exercise 4

Derive the transfer function of a zero-order hold (ZOH) device